Final Refined Modules

Objective

Summarize all improvements implemented during the final stabilization phase.

Refined Components

Stable Scoring Engine

- Safe score normalization
- Overflow prevention
- Consistent aggregation

Error Handling

- Exception wrapper
- Safe fallback responses
- Input validation

Stable API Layer

- Uniform API response format
- Consistent success/error fields

Conversation Logic

- Retry limit
- Clarification handling
- Skip logic

Edge Case Handling

- Empty responses
- Short responses
- Excessively long inputs
- Missing values

Data Validation

- Null value handling
- Safe numerical conversion

- Score boundary enforcement

Advantages

- Reduced crashes
- Consistent outputs
- Improved maintainability
- Better production readiness

Limitations

- Rule-based logic
- Requires continuous monitoring

Future Improvements

- AI anomaly detection
- Automatic debugging
- Live monitoring dashboard
- Self-healing pipelines